

Lecture 17: Intro to the Term Structure

Raul Riva

FGV EPGE

March, 2026

Intro

- Last lecture: modern asset pricing in high dimensions
- Today: move to the term structure of interest rates
- Goal: build the bond math, the EH benchmark, the classic evidence, and the Diebold-Li forecasting view
- This is an intro lecture, but not a shy one

Why This Lecture?

- Bond markets give us a laboratory for expected returns, discount rates, and macro expectations
- The yield curve is a cross section of prices every month
- So we need notation for both the cross section and the dynamics
- Main question: when the curve moves, is it expected short rates or term premia?

- Part I: bond yields, forward rates, and holding returns
- Part II: the expectations hypothesis as a benchmark
- Part III: what Fama and Bliss ([1987](#)) and Campbell and Shiller ([1991](#)) found
- Part IV: Nelson-Siegel and why Diebold and Li ([2006](#)) made it useful for forecasting

Part I: Building Blocks

Zero-Coupon Bonds and Yields

- $P_t^{(n)}$ be the time- t price of an n -period zero-coupon bond ([Gürkaynak and Wright 2012](#))
- It pays \$1 at time $t + n$ (we are abstracting away from defaults)

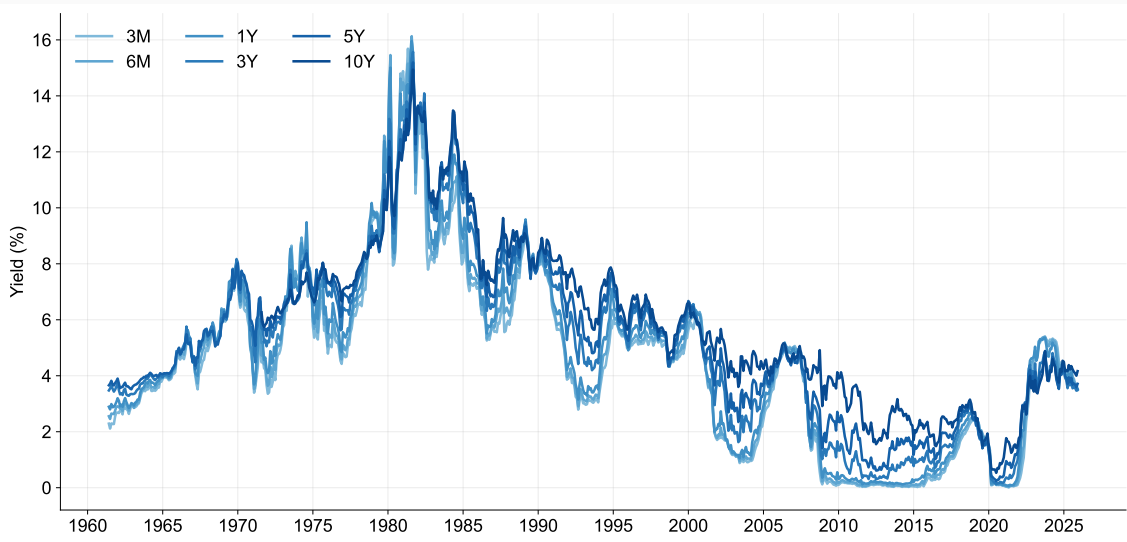
$$P_t^{(n)} = e^{-ny_t^{(n)}}, \quad y_t^{(n)} = -\frac{1}{n} \log P_t^{(n)}$$

- $y_t^{(n)}$ is the annualized continuously compounded yield
- Think about zero-coupon bonds are Arrow securities – they are our building block
- In most cases these yields are *not* observed directly. Why?

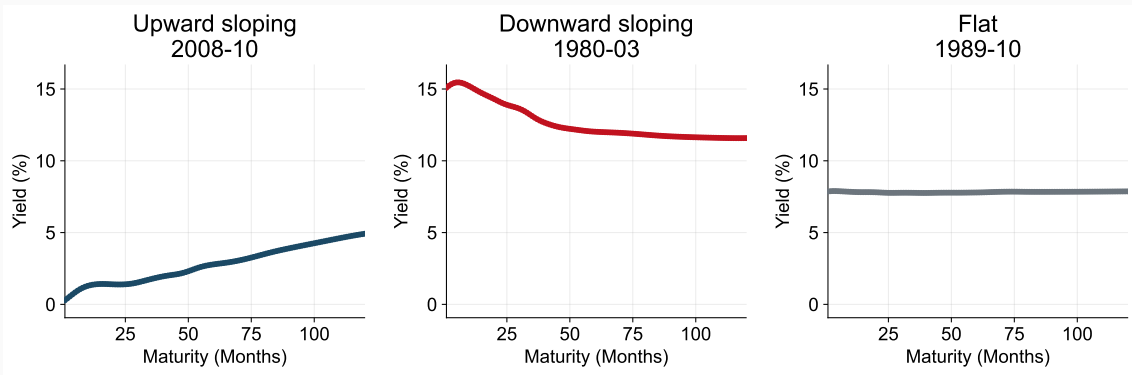
The Yield Curve

- The yield curve is the mapping $n \mapsto y_t^{(n)}$
- Same date, many maturities
- Long and short yields need not move one-for-one
- The curve can be upward sloping, downward sloping, flat, humped, ...
- There is a huge literature of how yields move over time and across n

US Treasury Yields Over Time



Yield Curve on Different Days



- The yield curve can look very different on different days
- Most frequently: upward sloping, and concave

Forward Rates

- Suppose you want to invest money for 2 years
- Option 1: buy a 2-year bond and get $y_t^{(2)}$ for the whole period
- Option 2: buy a 1-year bond, then reinvest the proceeds for another year

Forward Rates

- Suppose you want to invest money for 2 years
- Option 1: buy a 2-year bond and get $y_t^{(2)}$ for the whole period
- Option 2: buy a 1-year bond, then reinvest the proceeds for another year
- Return with option 1: $2 \cdot y_t^{(2)}$
- Return with option 2: $y_t^{(1)} + y_{t+1}^{(1)}$
- The return for option 2 is *random*, of course

Forward Rates

- Suppose you want to invest money for 2 years
- Option 1: buy a 2-year bond and get $y_t^{(2)}$ for the whole period
- Option 2: buy a 1-year bond, then reinvest the proceeds for another year
- Return with option 1: $2 \cdot y_t^{(2)}$
- Return with option 2: $y_t^{(1)} + y_{t+1}^{(1)}$
- The return for option 2 is *random*, of course
- The 1-year forward rate between $t + 1$ and $t + 2$ is the rate that makes the two options have the same expected return

$$f_t^{(1,1)} = 2y_t^{(2)} - y_t^{(1)}$$

Forward Rates:

- In general, the forward rate for an m -period investment starting n periods ahead is

$$f_t^{(n,m)} = \frac{1}{m} \left[(n+m)y_t^{(n+m)} - ny_t^{(n)} \right]$$

- When $n = 0$ and $m = 1$, this reduces to the short rate $y_t^{(1)}$
- Forward rates carry the same information as yields. Why?
- When $m \rightarrow 0$, we call it the instantaneous forward rate, $f_t^{(n)}$
- This is the “slope” of the yield curve at date t around maturity n

Yield = Average Forward Rates

For discrete one-year forwards,

$$y_t^{(n)} = \frac{1}{n} \sum_{i=1}^n f_t^{(i-1,1)}$$

Yield = Average Forward Rates

For discrete one-year forwards,

$$y_t^{(n)} = \frac{1}{n} \sum_{i=1}^n f_t^{(i-1,1)}$$

- A long yield is an average of future forward rates
- So the yield curve is already a summary of expected future financing conditions
- But it may also embed risk premia
- Notice that the equation above is a *tautology*

Holding-Period Returns

- Buy an n -period bond at t
- Sell it one period later as an $(n - 1)$ -period bond

$$r_{t+1}^{(n)} = \log P_{t+1}^{(n-1)} - \log P_t^{(n)} = ny_t^{(n)} - (n-1)y_{t+1}^{(n-1)}$$

- This is the realized one-period log return ([Gürkaynak and Wright 2012](#))
- Of course, we are abstracting away from taxes, transaction costs, and other frictions
- This return is random because $y_{t+1}^{(n-1)}$ is random

$$xr_{t+1}^{(n)} = r_{t+1}^{(n)} - y_t^{(1)} = ny_t^{(n)} - (n-1)y_{t+1}^{(n-1)} - y_t^{(1)}$$

- Compare the long bond with the risk-free one-period investment
- If $xr_{t+1}^{(n)} > 0$, the long bond beat the short rate over that period
- Empirical work often studies these excess returns directly
- Whenever people talk about “bond risk premia”, they usually mean expected excess returns

Part II: The Expectations Hypothesis

Notation Convention

- We will keep the same notation throughout the rest of the lecture
- $y_t^{(n)}$: yield at time t with n periods to maturity
- $r_{t+1}^{(n)}$: realized one-period return from buying an n -period bond at t
- $xr_{t+1}^{(n)} = r_{t+1}^{(n)} - y_t^{(1)}$
- $f_t^{(n,m)}$: forward rate for an m -period investment beginning n periods ahead

- The expectations hypothesis says long yields reflect expected future short yields plus a maturity-specific constant

$$y_t^{(n)} = \frac{1}{n} \sum_{i=0}^{n-1} E_t(y_{t+i}^{(1)}) + c_n$$

- If c_n is constant, time variation in yield spreads should mainly reflect changing expectations of future short rates

Log EH: One-Period Holding Return

In log form, EH can be written as

$$E_t \left[x r_{t+1}^{(n)} \right] = \mu_n$$

- Expected excess log return on an n -period bond is constant
- EH rules out time-varying expected bond premia
- This is why return predictability is so central

The same idea can be written as

$$E_t \left[y_t^{(n)} - \frac{1}{n} \sum_{i=0}^{n-1} y_{t+i}^{(1)} \right] = \theta_n$$

- Current long yields should line up with average future short yields, up to a constant
- Same restriction, different object

Under EH, spreads and forwards should forecast future short-rate movements:

$$y_t^{(2m)} - y_t^{(m)} \approx E_t \left[y_{t+m}^{(m)} - y_t^{(m)} \right] + c_{2m,m}$$

- High forward rates should mean high future short rates
- Not high expected excess returns

Once expected excess returns move over time,

$$E_t \left[x r_{t+1}^{(n)} \right] \neq \mu_n$$

- Then spreads and forward rates may forecast returns, not just rates
- That is why the empirical literature tests returns and spreads rather than average yields alone
- EH is a clean benchmark because it gives coefficients and signs to test

Questions?

Part III: Early Empirical Tests

Why These Regressions?

- EH is a conditional statement, so we test conditional forecasts
- Project future yield changes or excess returns on today's spreads
- Benchmark under EH: slope coefficients should be close to 1

$$y_{t+m}^{(m)} - y_t^{(m)} = a + b (y_t^{(n)} - y_t^{(m)}) + u_{t+m}, \quad xr_{t+1}^{(n)} = a + b [f_t^{(n-1,1)} - y_t^{(1)}] + u_{t+1}$$

- If b is far from 1, the EH benchmark is in trouble

Fama and Bliss (1987) focus on the forward-spot spread:

$$f_t^{(n-1,1)} - y_t^{(1)} = E_t \left[y_{t+n-1}^{(1)} - y_t^{(1)} \right] + E_t \left[x r_{t+1}^{(n)} \right]$$

Fama and Bliss (1987) focus on the forward-spot spread:

$$f_t^{(n-1,1)} - y_t^{(1)} = E_t \left[y_{t+n-1}^{(1)} - y_t^{(1)} \right] + E_t \left[x r_{t+1}^{(n)} \right]$$

- It can be decomposed into expected future short-rate changes
- And expected excess returns on longer bonds
- Their regressions ask: which margin actually moves?

- Forward rates forecast excess holding-period returns much better than future short-rate changes ([Fama and Bliss 1987](#))
- In their complementary regressions, slopes are close to 1 for term-premium variation
- And close to 0 for yield-change forecasts
- Translation: forwards mostly move with expected bond returns, not just expected short rates

Fama-Bliss: Updated Figure/Table Goes Here

- Leave space for an updated replication
- Leave space for a figure on forward-spot spreads and excess returns

Campbell and Shiller (1991) study yield spreads directly

$$s_t^{(n,m)} \equiv y_t^{(n)} - y_t^{(m)}$$

- Under EH, the spread should forecast either future short-rate increases
- Or the capital loss on the long bond
- Those are just two sides of the same restriction

- The slope coefficients are far from the EH benchmark of 1 ([Campbell and Shiller 1991](#))
- Often they are too small, and sometimes even have the wrong sign
- Yield spreads do forecast something
- But much of that variation looks like predictable excess returns, not clean short-rate expectations

Campbell-Shiller: Updated Figure/Table Goes Here

- Leave space for updated coefficients
- Leave space for a table or a “perfect-foresight spread” figure

One Message from Both Papers

- Forward rates and yield spreads move too much to be pure forecasts of future short rates
- They forecast bond returns and time-varying term premia
- EH remains the right benchmark
- But it is a bad literal description of the data

Part IV: Nelson-Siegel and Diebold-Li

- We do not just want one regression per maturity
- We want a parsimonious model for the whole cross section of yields
- And then a simple model for its dynamics
- This is the problem Diebold and Li ([2006](#)) tackles directly

Start from a smooth forward-rate curve ([Nelson and Siegel 1987](#); [Diebold and Li 2006](#)):

$$f_t^{(\tau)} = \beta_{1t} + \beta_{2t}e^{-\lambda\tau} + \beta_{3t}\lambda\tau e^{-\lambda\tau}$$

- Three components
- One decay parameter λ

The Implied Yield Curve

Averaging forward rates gives

$$y_t^{(\tau)} = \beta_{1t} + \beta_{2t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_{3t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right)$$

- Same factors at every maturity
- Different maturities load differently on them

Why Level, Slope, Curvature?

- Loading on β_{1t} is 1 at all maturities
- Loading on β_{2t} starts at 1 and decays to 0
- Loading on β_{3t} starts at 0, rises, then decays
- These are exactly the shapes we call level, slope, and curvature

What Each Factor Does

- β_{1t} shifts the whole curve
- β_{2t} moves the short end more than the long end
- β_{3t} bends the middle of the curve
- A small number of factors captures the main co-movements in yields

- Fix λ to a convenient value
- Diebold and Li (2006) use $\lambda = 0.0609$
- Each month, estimate β_{1t} , β_{2t} , β_{3t} from the cross section of yields
- Then treat those estimated coefficients as factor time series
- Much easier to interpret than an unrestricted factor model

Forecast the Factors, Forecast the Curve

Let the factors follow simple dynamics:

$$\beta_t = c + A\beta_{t-1} + u_t$$

- Forecast the factors with AR or VAR models
- Plug factor forecasts into the Nelson-Siegel loadings
- Result: a forecasting model for the entire curve ([Diebold and Li 2006](#))
- Key move: turn a cross-sectional decomposition into a dynamic forecasting system

Questions?

- Part I gave us the language: yields, forwards, and holding returns
- Part II gave us the benchmark: EH says spreads should reflect expected short rates, not time-varying premia
- Part III showed the classic failure: forwards and spreads forecast returns too well
- Part IV gave a practical alternative: summarize the whole curve with level, slope, and curvature, then forecast those factors
- The term structure is about both prices today and expected returns tomorrow

Readings and References i

Campbell, John Y., and Robert J. Shiller. 1991. "Yield Spreads and Interest Rate Movements: A Bird's Eye View." *The Review of Economic Studies* 58 (3): 495–514.

Diebold, Francis X., and Canlin Li. 2006. "Forecasting the Term Structure of Government Bond Yields." *Journal of Econometrics* 130 (2): 337–64.

Fama, Eugene F., and Robert R. Bliss. 1987. "The Information in Long-Maturity Forward Rates." *The American Economic Review* 77 (4): 680–92.

Gürkaynak, Refet S., and Jonathan H. Wright. 2012. "Macroeconomics and the Term Structure." *Journal of Economic Literature* 50 (2): 331–67.

Nelson, Charles R., and Andrew F. Siegel. 1987. "Parsimonious Modeling of Yield Curves." *The Journal of Business* 60 (4): 473–89.